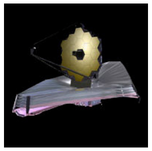

12435 - Massive stars in the first gigayear revealed with an ultra-deep NIRSpec stare

Cycle: 5, Proposal Category: GO

INVESTIGATORS

Name	***Institution***
Dr. Peter Senchyna (PI)	**Carnegie Institution of Washington**
Prof. Daniel P. Stark (CoI)	University of California - Berkeley
Dr. Zuyi Chen (CoI) (ESA Member)	University of Copenhagen, Niels Bohr Institute
Prof. Charlotte Mason (CoI) (ESA Member)	University of Copenhagen, Niels Bohr Institute
Dr. Mengtao Tang (CoI)	Tsung-Dao Lee Institute, Shanghai Jiao Tong University
Dr. Lily Whitler (CoI) (ESA Member)	University of Cambridge, Kavli Institute for Cosmology
Dr. Stephane Charlot (CoI) (ESA Member)	CNRS, Institut d'Astrophysique de Paris
Adele Plat (CoI) (ESA Member)	Ecole Polytechnique Federale de Lausanne
Dr. Keerthi Vasan G.C. (CoI)	Carnegie Institution of Washington
Dr. Grace Olivier (CoI)	Carnegie Institution of Washington
Lauren Henson (CoI)	Carnegie Institution of Washington

OBSERVATIONS

Folder	***Observation***	***Label***	***Observing Template***	***Science Target***
Observation Folder				
	1	a2744_G140H	NIRSpec MultiObject Spectroscopy	(2) a2744_cat_v1

ABSTRACT

JWST has transformed our view of galaxies at z>6 and in the process revealed dramatic mysteries: including surprisingly UV-bright galaxies, systems host to highly ionized and nitrogen enriched gas, and enigmatic little red dots. Underlying all of these mysteries is a central theme: hints that massive star populations in the early Universe are fundamentally different from even those in the best analogs at low redshift. The physics of star

formation under high-density early Universe conditions might be expected to yield such differences: with an extremely high fraction of massive stars born in dense cluster conditions unique to these early times. The potential impact of stellar populations and an IMF impacted by such processes are large, resulting in substantial systematic uncertainties in interpreting integrated light observations of all sorts at high-z. Here we propose a program designed to directly confront these uncertainties: an extraordinarily deep single-grating stare in G140H to construct the highest-quality rest-UV spectroscopy possible with JWST for a statistical sample of z=7-9 galaxies. With these spectra we wil directly constrain stellar wind and (in the brightest systems) photospheric features, sensitive both to the abundances and ages of the stellar populations present and to the presence of very massive stars. Together with unprecedentedly deep constraints on UV emission in several LRDs and on high-ionization UV line emission in a large sample of faint z=5-9 systems observed on the same MSA, these data will allow us to asssemble a sorely-needed first detailed picture of massive stars and their impacts in the reionization era.

OBSERVING DESCRIPTION

Our program is built around deep spectroscopy of a sample of lensed z=7-9 galaxies in Abell-2744. Our science there is dependent upon achieving very high SNR for these galaxies in the restframe far-ultraviolet continuum. To achieve this, we request 110 hours science exposure, corresponding to a total charged time of 137 hours.

The G140H grating uniquely satisfies our primary science wavelength range (1240-1640A rest-frame) and spectral resolution (R>2000) requirements. The total requested exposure time is based upon ETC calculations for simulated stellar populations with a range of properties as described in the proposal. Our simulations suggest that recovering stellar wind feature at high-confidence for a metal-poor stellar populations without additional sources like very massive stars will require SNR>10; and a resulting total integration time of 110 hours at the AB magnitudes of our primary four bright targets.

Our goal is to maximize on-source exposure and SNR while minimizing overheads. The ETC indicates that the observations are not background limited or in danger of saturation. Data volume considerations from the APT necessitate that we utilize NRSIRS2 as opposed to NRSIRS2RAPID. We plan 3-point nod-in-slitlet observations with a standard 3 shutter slitlet to maximize SNR.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	a2744_m26	RA: 00 14 26.0427 (3.6085113d) Dec: -30 24 22.90 (-30.40636d) Equinox: J2000		
Comments: *Description=[]*				
(2)	a2744_cat_v1	RA: 00 14 22.3021 (3.5929254d) Dec: -30 24 9.80 (-30.40272d) Equinox: J2000		
Comments: *Description=[]*				
(3)	a2744_m26_wl	RA: 00 14 26.0427 (3.6085113d) Dec: -30 24 22.90 (-30.40636d) Equinox: J2000		
Comments: *Description=[]*				

Observation

Proposal 12435, Observation 1: a2744_G140H

Fri Mar 13 23:04:41 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec MultiObject Spectroscopy

Diagnostics

(a2744_G140H (Obs 1)) Warning (Form): Config c1 (#1) has 1 filler slits affected by failed closed shutters.
(a2744_G140H (Obs 1)) Warning (Form): Config c1 (#10) has 1 filler slits affected by failed closed shutters.
(a2744_G140H (Obs 1)) Warning (Form): Config c1 (#11) has 1 filler slits affected by failed closed shutters.
(a2744_G140H (Obs 1)) Warning (Form): Config c1 (#12) has 1 filler slits affected by failed closed shutters.
(a2744_G140H (Obs 1)) Warning (Form): Config c1 (#13) has 1 filler slits affected by failed closed shutters.
(a2744_G140H (Obs 1)) Warning (Form): Config c1 (#14) has 1 filler slits affected by failed closed shutters.
(a2744_G140H (Obs 1)) Warning (Form): Config c1 (#2) has 1 filler slits affected by failed closed shutters.
(a2744_G140H (Obs 1)) Warning (Form): Config c1 (#3) has 1 filler slits affected by failed closed shutters.
(a2744_G140H (Obs 1)) Warning (Form): Config c1 (#4) has 1 filler slits affected by failed closed shutters.
(a2744_G140H (Obs 1)) Warning (Form): Config c1 (#5) has 1 filler slits affected by failed closed shutters.
(a2744_G140H (Obs 1)) Warning (Form): Config c1 (#6) has 1 filler slits affected by failed closed shutters.
(a2744_G140H (Obs 1)) Warning (Form): Config c1 (#7) has 1 filler slits affected by failed closed shutters.
(a2744_G140H (Obs 1)) Warning (Form): Config c1 (#8) has 1 filler slits affected by failed closed shutters.
(a2744_G140H (Obs 1)) Warning (Form): Config c1 (#9) has 1 filler slits affected by failed closed shutters.
(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:7) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	a2744_cat_v1	RA: 00 14 22.3021 (3.5929254d) Dec: -30 24 9.80 (-30.40272d) Equinox: J2000		

Comments:
Description=[]

Proposal 12435 - Observation 1 - Massive stars in the first gigayear revealed with an ultra-deep NIRSpec stare

Acquisition

#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
2		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
3		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
4		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
5		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
6		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
7		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	

Template

TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold
MSATA	false	No	MSA Center	primary (239 sources)	filler (4251 sources)	jwst-nirspec-g140h	1.5

Reference Stars

Spectral Elements

#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
1	1 (G140H/F100LP)	c1	3 Shutter Slitlet	3.6098775 Degrees - 30.401111111111106 Degrees	29.990715166684744			3	15	28667.169
2	1 (G140H/F100LP)	c1	3 Shutter Slitlet	3.6098775 Degrees - 30.401111111111106 Degrees	29.990715166684744			3	15	28667.169
3	1 (G140H/F100LP)	c1	3 Shutter Slitlet	3.6098775 Degrees - 30.401111111111106 Degrees	29.990715166684744			3	15	28667.169
4	1 (G140H/F100LP)	c1	3 Shutter Slitlet	3.6098775 Degrees - 30.401111111111106 Degrees	29.990715166684744			3	15	28667.169
5	1 (G140H/F100LP)	c1	3 Shutter Slitlet	3.6098775 Degrees - 30.401111111111106 Degrees	29.990715166684744			3	15	28667.169
6	1 (G140H/F100LP)	c1	3 Shutter Slitlet	3.6098775 Degrees - 30.401111111111106 Degrees	29.990715166684744			3	15	28667.169
7	1 (G140H/F100LP)	c1	3 Shutter Slitlet	3.6098775 Degrees - 30.401111111111106 Degrees	29.990715166684744			3	15	28667.169
8	1 (G140H/F100LP)	c1	3 Shutter Slitlet	3.6098775 Degrees - 30.401111111111106 Degrees	29.990715166684744			3	15	28667.169
9	1 (G140H/F100LP)	c1	3 Shutter Slitlet	3.6098775 Degrees - 30.401111111111106 Degrees	29.990715166684744			3	15	28667.169
10	1 (G140H/F100LP)	c1	3 Shutter Slitlet	3.6098775 Degrees - 30.401111111111106 Degrees	29.990715166684744			3	15	28667.169
11	1 (G140H/F100LP)	c1	3 Shutter Slitlet	3.6098775 Degrees - 30.401111111111106 Degrees	29.990715166684744			3	15	28667.169
12	1 (G140H/F100LP)	c1	3 Shutter Slitlet	3.6098775 Degrees - 30.401111111111106 Degrees	29.990715166684744			3	15	28667.169
13	1 (G140H/F100LP)	c1	3 Shutter Slitlet	3.6098775 Degrees - 30.401111111111106 Degrees	29.990715166684744			3	15	28667.169
14	1 (G140H/F100LP)	c1	3 Shutter Slitlet	3.6098775 Degrees - 30.401111111111106 Degrees	29.990715166684744			3	15	28667.169

Special Requirements	
	Group Visits within 53.0 Days Visits Same PA MSA Planned Aperture PA 29.9993 to 29.9993 Degrees (V3 251.4247303 to 251.4247303)